



DE ZG535 (23-S2) - Sessions 3 - 5 (27 Jan, 3 & 10 Feb 2024)

BITS WILP M Tech Data Science & Engineering (Birla Institute of Technology and Science, Pilani)



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Sessions 3 - 5: Bases and dimension of a subspace

Topics: Linear independence; Bases and dimension of a subspace; Bases for the column space and null space of a matrix and its rank and nullity; Existence and uniqueness of the solution of a system of linear equations
MATLAB demo: Bases for the column space and null space of a matrix, rank and nullity of a matrix

Introduction

We shall introduce the concept of linear independence of the columns of a matrix, followed by the rank of a matrix and its significance in the nature of the solution of a system of m linear equations in n unknowns. The bases and dimension of a subspace are then addressed. The bases for the column space and null space of a matrix are considered next. Finally, the applications of these concepts in the existence and uniqueness of the solution of linear equations are explored.

Linear independence

The set $S = \{u_1, u_2, u_3 \dots u_n\}$ of n vectors is said to be linearly dependent if scalars $c_1, c_2, c_3 \dots c_n$, not all zero (at least one is nonzero), exist, such that $c_1u_1 + c_2u_2 + \dots + c_nu_n = 0$. If the scalars $c_1, c_2, c_3 \dots c_n$ being all zero is the only way to satisfy $c_1u_1 + c_2u_2 + \dots + c_nu_n = 0$, the set S is said to be linearly independent.

If S is linearly dependent, then some c_k in $c_1u_1 + c_2u_2 + \dots + c_ku_k + \dots + c_nu_n = 0$ will be a non-zero scalar. Then we can write:

$$u_k = -\frac{c_1}{c_k}u_1 - \frac{c_2}{c_k}u_2 - \dots - \frac{c_n}{c_k}u_n$$

i.e.; if the set $S = \{u_1, u_2, u_3 \dots u_n\}$ of n vectors is linearly dependent, at least one of the vectors can be expressed as a linear combination of the rest.

Note: If the set S includes the null vector (say u_1), then $c_1u_1 + 0u_2 + 0u_3 \dots + 0u_n = 0$ for any $c_1 \neq 0$. The set S is linearly dependent as per the above definition.

Example 1: Show that $u_1 = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$, $u_2 = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}$ and $u_3 = \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix}$ are linearly dependent.

Express u_1 as a linear combination of u_2 and u_3 .

$$c_1 u_1 + c_2 u_2 + c_3 u_3 = \begin{bmatrix} 1 & 1 & 1 \\ 3 & -2 & 4 \\ -1 & 4 & -2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ must have a non-zero solution.}$$

Obtain the row-echelon form of the augmented matrix as:

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 3 & -2 & 4 & 0 \\ -1 & 4 & -2 & 0 \end{bmatrix} \xrightarrow{\begin{matrix} -3R_1+R_2 \rightarrow R_2 \\ R_1+R_3 \rightarrow R_3 \end{matrix}} \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & -5 & 1 & 0 \\ 0 & 5 & -1 & 0 \end{bmatrix} \xrightarrow{R_2+R_3 \rightarrow R_3} \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & -5 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Here, let the free variable $c_3 = t$. Then, $c_2 = \frac{1}{5}t$ and $c_1 = -\frac{6}{5}t$

Hence, $c_1 = -6$, $c_2 = 1$ and $c_3 = 5$ is a solution.

$$\text{i.e.; } -6u_1 + u_2 + 5u_3 = 0 \Rightarrow u_1 = \frac{1}{6}u_2 + \frac{5}{6}u_3 \quad \blacksquare$$

Bases and dimension of a subspace

We have seen that the span of a set S of vectors is the subspace that consists of all the linear combinations of S . The set S may not be independent, which means that the dependent vectors in S may be removed and the leftover elements of S still span the same subspace. i.e.; the smallest set of vectors whose linear combination covers the entire subspace is called the basis of the subspace.

The basis of the subspace R is the set S of linearly independent vectors whose span is R . Clearly, S represents the minimal spanning set and is not unique (every subspace R except the null subspace has infinitely many bases) for R . For a given S , any vector in R is uniquely represented as the linear combination of the elements of S . The number of elements of S is called the dimension of the subspace R .

While a subspace contains infinitely many vectors, its basis is a finite set. The basis of a subspace of R^n consists of n vectors at most.

Bases for the column space and null space of a matrix and its rank and nullity

A basis for the column space of a matrix is its columns corresponding to the leading variables in its row-echelon form.

To obtain a basis for the null space of a matrix, augment a column of zeros to the coefficient matrix and obtain the solution as a linear combination of vectors. Those vectors form a basis for the null space of the matrix.

The **rank** and **nullity (degeneracy or rank deficiency)** of a matrix are the dimensions of its column space and null space respectively. It can be shown that the sum of the rank and nullity of an $m \times n$ matrix is n .

Example 2: Find bases for the column space and null space of $A = \begin{bmatrix} 1 & 3 & 4 & 3 \\ 3 & 9 & 12 & 3 \\ 1 & 3 & 4 & 1 \end{bmatrix}$ and

hence determine its rank and nullity. Verify using the MATLAB command `colspace(A)`. Does the MATLAB output match the calculations? Why?

Obtain the row-echelon form of A as:

$$\begin{bmatrix} 1 & 3 & 4 & 3 \\ 3 & 9 & 12 & 3 \\ 1 & 3 & 4 & 1 \end{bmatrix} \xrightarrow{\substack{-3R_1+R_2 \rightarrow R_2 \\ -R_1+R_3 \rightarrow R_3}} \begin{bmatrix} 1 & 3 & 4 & 3 \\ 0 & 0 & 0 & -6 \\ 0 & 0 & 0 & -2 \end{bmatrix} \xrightarrow{-\frac{1}{6}R_2 \rightarrow R_2} \begin{bmatrix} 1 & 3 & 4 & 3 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -2 \end{bmatrix} \xrightarrow{2R_2+R_3 \rightarrow R_3} \begin{bmatrix} 1 & 3 & 4 & 3 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The column space of dimension 2 has the set of basis vectors $B = \left\{ \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 3 \\ 1 \end{bmatrix} \right\}$.

Rank of A is 2 (dimension of its column space)

Obtain the basis for the null space from the row-echelon form of the matrix A augmented with a column of zeros. This will be the same as the matrix found above augmented with a column of zeros. i.e.;

$$\begin{bmatrix} 1 & 3 & 4 & 3 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$x_3 = t$ and $x_2 = s$ are the free variables. Then, $x_4 = 0$ and $x_1 = -3s - 4t$

$$\text{i.e.}; \begin{bmatrix} -3s - 4t \\ s \\ t \\ 0 \end{bmatrix} = s \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -4 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

The null space of dimension 2 has the set of basis vectors $B = \left\{ \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -4 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}$.

Nullity of A is 2 (dimension of its null space).

Verify that rank + nullity = n ($=4$)

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Existence and uniqueness of the solution of a system of linear equations

We use the concepts of column space, null space, rank and nullity in establishing the existence and uniqueness of the solution of a system of linear equations.

Existence of the solution of $Ax = b$ is established as follows: In the last session, we saw that a solution of $Ax = b$ exists (at least one solution) if and only if b is in the column space of A . If a solution must exist for every b , the columns of A must span R^m , which requires the dimension of the column space of A (= "rank of A ") to be m ($m \leq n$ since the rank of A cannot exceed the smaller of m and n).

Explanation of the above with a system of 3 equations in 8 unknowns:

Case 1: Solution doesn't exist. If solution exists, it is not unique

$$\begin{bmatrix} 1 & -1 & 2 & 0 & -2 & -3 & 4 & 6 \\ 2 & -2 & 4 & 0 & -4 & -6 & 8 & 12 \\ 3 & -3 & 6 & 1 & -6 & -9 & 12 & 18 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \\ x_7 \\ x_8 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix} \qquad \begin{bmatrix} 1 & -1 & 2 & 0 & -2 & -3 & 4 & 6 \\ 2 & -2 & 4 & 0 & -4 & -6 & 8 & 12 \\ 3 & -3 & 6 & 1 & -6 & -9 & 12 & 18 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \\ x_7 \\ x_8 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$$

Case 2: Solution must exist for all b

$$\begin{bmatrix} 1 & 4 & 0 & 0 & -2 & -3 & 4 & 6 \\ 2 & 0 & 6 & 0 & -4 & -6 & 8 & 12 \\ 3 & 0 & 0 & 1 & -6 & -9 & 12 & 18 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \\ x_7 \\ x_8 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

The question of the uniqueness of the solution of $Ax = b$, which arises only if a solution exists, is related to the dimension of the null space of A . For $Ax = b$ to have a unique solution (provided a solution exists), the nullity of A must be zero – i.e.; the null space of A must contain just the zero vector.

If $Ax = b$ is to have a unique solution for every b , we should be able to express every b as a unique linear combination of the columns of A , which means that the columns of A must be linearly independent (i.e.; the columns of A must be a basis set for the column space of A) or the rank of A must be n (and $n \leq m$)

Let the nullity of A be non-zero, and let y belong to the null space of A (i.e.; $Ay = 0$). If x is a solution of $Ax = b$, so is $x + y$; since $A(x + y) = Ax + Ay = b + 0 = b$. i.e.; if the solution of $Ax = b$ exists, it is not unique.

Example 3: Examine the nature of the solutions of the system $Ax = b$ of 4 equations in 4 unknowns if the rank of A is 3.

If the rank of A is 3, the column space of A cannot span the whole of R^4 . This means the system will be inconsistent for certain b vectors. For other b vectors that are spanned by the column space of A , the system will be consistent with infinite solutions (because the nullity of A is $4 - 3 = 1$). ■

We may summarize the above as follows:

- (i) If the ranks of A and the augmented matrix are unequal, the system is inconsistent
- (ii) If the ranks of A and the augmented matrix are equal, the system is consistent – if both are n , the solution is unique; if both are less than n , there are infinitely many solutions

In-class practice: Examine the nature of the solution in the following three cases:

(i) $x_1 + x_3 = 3, \quad x_2 + x_3 = -1, \quad x_1 + x_3 = 5$

(ii) $x_1 + 2x_2 = 2, \quad 3x_1 + 4x_2 = 3, \quad 5x_1 + 6x_2 = 4$

(iii) $x_1 + 3x_2 + 5x_3 = 3, \quad 2x_1 + 8x_2 + 12x_3 = 7, \quad -x_1 + 5x_2 + 3x_3 = 1,$
 $-2x_1 + 8x_2 + 4x_3 = 1, \quad x_1 + 3x_2 + 5x_3 = 3$

Take 10 minutes to do this problem (solution on next page)

SOLUTION:

$$(i) \quad \begin{bmatrix} 1 & 0 & 1 & 3 \\ 0 & 1 & 1 & -1 \\ 1 & 0 & 1 & 5 \end{bmatrix} \xrightarrow{-R_1+R_3 \rightarrow R_3} \begin{bmatrix} 1 & 0 & 1 & 3 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

rank (A) = 2, rank (aug-A) = 3. No solution.

$$(ii) \quad \begin{bmatrix} 1 & 2 & 2 \\ 3 & 4 & 3 \\ 5 & 6 & 4 \end{bmatrix} \xrightarrow{\begin{matrix} -3R_1+R_2 \rightarrow R_2 \\ -5R_1+R_3 \rightarrow R_3 \end{matrix}} \begin{bmatrix} 1 & 2 & 2 \\ 0 & -2 & -3 \\ 0 & -4 & -6 \end{bmatrix} \xrightarrow{-2R_2+R_3 \rightarrow R_3} \begin{bmatrix} 1 & 2 & 2 \\ 0 & -2 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

rank (A) = 2, rank (aug-A) = 2. Both are n .

Unique solution (check $x_1 = -1, x_2 = \frac{3}{2}$).

$$(iii) \quad \begin{bmatrix} 1 & 3 & 5 & 3 \\ 2 & 8 & 12 & 7 \\ -1 & 5 & 3 & 1 \\ -2 & 8 & 4 & 1 \\ 1 & 3 & 5 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 5 & 3 \\ 0 & 2 & 2 & 1 \\ 0 & 8 & 8 & 4 \\ 0 & 14 & 14 & 7 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 5 & 3 \\ 0 & 2 & 2 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

rank (A) = 2, rank (aug-A) = 2. Both are less than n (3). Infinitely many solutions.

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